

Vulnerability Assessment Report

6th September 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 20XX to August 20XX. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

Conducting this vulnerability assessment is crucial because the database server holds vital customer information that drives the e-commerce business. Securing this data is essential to maintain customer trust, prevent regulatory fines, and avoid severe financial loss. If the server were disabled or breached, business operations would halt completely, leading to massive reputational damage.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
<i>E.g. Competitor</i>	<i>Obtain sensitive information via exfiltration</i>	1	3	3
<i>External Hacker</i>	<i>Denial of Service (DoS) attack on the database</i>	2	3	6
<i>Insider Threat</i>	<i>Unauthorised data modification or deletion</i>	1	2	2

Approach

Risks considered the data storage and management methods of the business. The likelihood of a threat occurrence and the impact of these potential events were weighed against the risks to day-to-day operational needs.

Remediation Strategy

Implementation of authentication, authorization, and auditing mechanisms to ensure that only authorized users access the database server. This includes using strong passwords, role-based access controls, and multi-factor authentication to limit user privileges. Encryption of data in motion using TLS instead of SSL. IP allow-listing to corporate offices to prevent random users from the internet from connecting to the database.